

Modification of Neural Circuit Functions by Microglial P2Y6 Receptors in Health and Neurodegeneration.

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Neural circuits consisting of neurons and glial cells help to establish all functions of the CNS. Microglia, the resident immunocytes of the CNS, are endowed with UDP-sensitive P2Y6 receptors (P2Y6Rs) which regulate phagocytosis/pruning of excessive synapses during individual development and refine synapses in an activity-dependent manner during adulthood. In addition, this type of receptor plays a decisive role in primary (Alzheimer's disease, Parkinson's disease, neuropathic pain) and secondary (epilepsy, ischemic-, mechanical-, or irradiation-induced) neurodegeneration. A whole range of microglial cytokines controlled by P2Y6Rs, such as the interleukins IL-1 β , IL-6, IL-8, and tumor necrosis factor- α (TNF- α), leads to neuroinflammation, resulting in neurodegeneration. Hence, small molecular antagonists of P2Y6Rs and genetic knockdown of this receptor provide feasible ways to alleviate inflammation-induced neurological disorders but might also interfere with the regulation of the synaptic circuitry. The present review aims at investigating this dual role of P2Y6Rs in microglia, both in shaping neural circuits by targeted phagocytosis and promoting neurodegenerative illnesses by fostering neuroinflammation through multiple transduction mechanisms.

A Distinct Population of Microglia Supports Adult Neurogenesis in the Subventricular Zone.

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Microglia are involved in synaptic pruning both in development and in the mature CNS. In this study, we investigated whether microglia might further contribute to circuit plasticity by modulating neuronal recruitment from the neurogenic subventricular zone (SVZ) of the adult mouse striatum. We found that microglia residing in the SVZ and adjacent rostral migratory stream (RMS) comprise a morphologically and antigenically distinct phenotype of immune effectors. Whereas exhibiting characteristics of alternatively activated microglia, the SVZ/RMS microglia were clearly distinguished by their low expression of purinoceptors and lack of ATP-elicitable chemotaxis. Furthermore, the *in vivo* depletion of these microglia hampered the survival and migration of newly generated neuroblasts through the RMS to the olfactory bulb. SVZ and RMS microglia thus appear to comprise a functionally distinct class that is selectively adapted to the support and direction of neuronal integration into the olfactory circuitry. Therefore, this unique microglial subpopulation may serve as a novel target with which to modulate cellular addition from endogenous neural stem and progenitor cells of the adult brain. Microglial cells are a specialized population of macrophages in the CNS, playing key roles as immune mediators. As integral components in the CNS, the microglia stand out for using the same mechanisms, phagocytosis and cytochemokine release, to promote homeostasis, synaptic pruning, and neural circuitry sculpture. Here, we addressed microglial functions in the subventricular zone (SVZ), the major postnatal neurogenic niche. Our results depict microglia as a conspicuous component of SVZ and its anterior extension, the rostral migratory stream, a pathway used by neuroblasts during their transit toward olfactory bulb layers. In addition to other unique populations residing in the SVZ niche, microglia display distinct morphofunctional properties that boost neuronal progenitor survival and migration in the mammalian brain.

Contribution of Neurons and Glial Cells to Complement-Mediated Synapse Removal during Development, Aging and in Alzheimer's Disease.

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Synapse loss is an early manifestation of pathology in Alzheimer's disease (AD) and is currently the best correlate to cognitive decline. Microglial cells are involved in synapse pruning during development via the complement pathway. Moreover, recent evidence points towards a key role played by glial cells in synapse loss during AD. However, further contribution of glial cells and the role of neurons to synapse pathology in AD remain not well understood. This review is aimed at comprehensively reporting the source and/or cellular localization in the CNS-in microglia, astrocytes, or neurons-of the triggering components (C1q, C3) of the classical complement pathway involved in synapse pruning in development, adulthood, and AD.

[Role of microglia in neuronal circuit formation].

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Microglia are known to maintain the cellular environment in the brain via their phagocytic activity. Further, recent studies have demonstrated that microglia play an important role in neuronal circuit formation. During neuronal development, new axonal and dendritic processes are formed, but excessive connections are eliminated appropriately; microglia participate in this pruning process. Moreover, evidence shows that microglia contribute to synaptic maturation by secreting various molecules.

Microglia: Phagocytosing to Clear, Sculpt, and Eliminate.

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Microglia are the primary phagocytes of the central nervous system. They eliminate excess functional connections between neurons to sculpt neuronal circuits during development and throughout adulthood. Understanding how microglia recognize and prune synapses during development is providing insight into synapse loss and dysfunction in disease.

Microglia: Lifelong modulator of neural circuits.

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Microglia, the sole immune cells in the brain, are the key player for synaptic regulation required for our brain function in both developing and adult brain. They have highly motile processes to detect synaptic functions. Recent accumulated studies have unveiled the mechanism underlying synapse detection and pruning / formation by microglia. In this review, we summarize the current knowledge of various microglial machinery recruited in synaptic modulation in the different life stages and contexts.

The brain's polymath: Emerging roles of microglia throughout brain development.

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Microglia, the resident brain immune cells, have garnered a reputation as major effectors of circuit wiring due to their ability to prune synapses. Other roles of microglia in regulating neuronal circuit development have so far received comparatively less attention. Here, we review the latest studies that have contributed to our increased understanding of how microglia regulate brain wiring beyond their role in synapse pruning. We summarize recent findings showing that microglia regulate neuronal numbers and influence neuronal connectivity through a bidirectional communication between microglia and neurons, processes regulated by neuronal activity and the remodeling of the extracellular matrix. Finally, we speculate on the potential contribution of microglia to the development of functional networks and propose an integrative view of microglia as active elements of neural circuits.

Microglia function during brain development: New insights from animal models.

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The role of microglia in healthy brains is just beginning to receive notice. Recent studies have revealed that these phagocytic cells control the patterning and wiring of the developing central nervous system (CNS) by regulating, amongst many other processes, programmed cell death, activity-dependent synaptic pruning and synapse maturation. Microglia also play important roles in the mature brain and have demonstrated effects on behavior. Converging evidence from human and mouse studies together raise questions as to the role of microglia in disorders of brain development such as autism and, schizophrenia. In this review, we summarize a number of major findings regarding the role of microglia in brain development and highlight some key questions and avenues for future study. This article is part of a Special Issue entitled SI: Neuroimmunology in Health And Disease.

Microglia in development: linking brain wiring to brain environment.

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Microglia are enigmatic non-neuronal cells that infiltrate and take up residence in the brain during development and are thought to perform a surveillance function. An established literature has documented how microglia are activated by pathogenic stimuli and how they contribute to and resolve injuries to the brain. However, much less work has been aimed at understanding their function in the uninjured brain. A series of recent in vivo imaging studies shows that microglia in their resting state are highly motile and actively survey their neuronal surroundings. Furthermore, new data suggest that microglia in their resting state are able to phagocytose unwanted synapses and in this way contribute to synaptic pruning and maturation during development. Coupled with their exquisite sensitivity to pathogenic stimuli, these data suggest that microglia form a link that couples changes in brain environment to changes in brain wiring. Here we discuss this hypothesis and propose a model for the role of microglia during development in sculpting brain connectivity.

Microglia: first responders in the central nervous system.

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Microglia has emerged not only as an essential inflammatory cell but also as a major player in the development of the adult brain. Microglia phagocytize extra-numerical synapses during postnatal development, maintain and strengthen the remaining subset of synapses, remodel synaptic circuits and clearing apoptotic newborn neurons. Thereby, microglia plays a crucial role for the establishment, plasticity and function of adult neural circuits. In addition to the key role in normal brain function, any imbalance in microglia activity has been associated with neurodegenerative diseases. Microglial cells respond rapidly to smallest pathological changes, this being a vital aspect in many tissue scaring and the local confinement of focal lesions. It is assumed that the high motility of microglial cells represents an important requirement to fulfill the numerous functions. In this review will highlight the role of microglial motility in the healthy and the injured brain, and discuss how impairment of microglia motility can affect normal brain function.
